

CS 679 Course Project Proposal

Physics-Informed Neural Networks for Option Pricing: Reproduction and Stability/Threshold Extensions

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Physics-Informed Neural Networks in Pricing Financial Options (Socaciu & Paşcu, 2025)

The paper uses a Physics-Informed Neural Networks (PINN) to solve the Black–Scholes partial differential equation (PDE) by embedding the PDE residual and boundary/terminal constraints into the loss function.

Objective

✓ Our objective is to reproduce the results and provide evidence that the method produces option prices consistent with benchmark solutions.

We aim to:

- Reproduce core pricing results for European options by solving the Black–Scholes PDE with a PINN.
- Validate accuracy against the closed-form Black–Scholes formula and/or a classical numerical baseline .
- Report quantitative error metrics and visualizations (pricing curves/surfaces) to demonstrate that the approach works as described.

Computational Considerations

PINN training can be expensive if using very large networks or dense collocation sampling. To ensure feasibility on typical local resources:

- Start with European options (1D spatial PDE) and a small network (e.g., 3–4 hidden layers).
- Use a limited number of collocation points (interior + boundary) and scale up only if needed.

Extension: Financial Variations, Stability, Thresholds, and Emergent Behavior

After reproducing some results, we will consider exploring one or more of the following extensions to better understand both the behavior of PINNs and their applicability to financial problems and beyond.

We may investigate how enforcing PDE structure affects the stability and robustness of the learned solution:

- Financial Variations: different payoff structures (instead of only European call, add European put) and/or compare different maturities / parameters.
- Sensitivity / robustness tests: vary initialization, collocation sampling density, and noise in sampling to assess stability of the learned pricing surface.
- Threshold-like training effects: explore whether there is a regime change where accuracy improves sharply once collocation points/epochs exceed a certain level (a “threshold” effect), and document this behavior empirically.

(Conceptual Interpretation) We can interpret these observations using ideas from stability of dynamical systems and complex systems (e.g., global structure emerging from local constraints), and discuss possible connections to broader topics such as hedging as feedback control and stability-inspired objectives (e.g., Lyapunov-style constraints) as future work.

one of these would be fine. :)